Experimental Design Final Project Report

Abstract

Hotel operators consistently face challenges in managing unpredictable booking cancellations, which thereby affect revenue management and operational planning. This study investigates whether the booking channel—online vs. offline—is a significant determinant of hotel booking cancellations.

Using observational data from two hotels in Portugal and applying a quasi-experimental design framework, we estimate the effect of booking channel on cancellation behavior after adjusting for key confounders. This dataset in question is observational. All relevant diagrams can be found in code output (given as an appendix with dataset)

Data Facts

Dataset used:**https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand**

The data was obtained from a public Kaggle repository and originally sourced from hotels' Property Management System (PMS) SQL databases. It includes booking records from two hotels in Portugal: a city hotel in Lisbon and a resort hotel in Algarve. The data contains both raw and preprocessed variables. Some factor variables used in the analysis were engineered from existing ones (for example, converting `market_segment` into a binary booking channel indicator).

Extensive information about data columns found here:
https://www.sciencedirect.com/science/article/pii/S2352340918315191

The objective metric utilized for this experiment is the cancellation rate, which is expressed as the binary variable is_canceled.

Null Hypothesis (H_0) = Cancellation rates are the same for Online and Offline bookings.

Alternative Hypothesis (H_a) = Cancellation rates are different for Online vs. Offline.

The key variable in question, the multi-categorical variable market_segment has been reclassified into a binary variable distinguishing between Online vs Offline channels. In this classification, only Online Travel Agencies (OTAs) are considered 'Online,' while all other segments are classified as 'Offline'.

This reclassified binary variable called booking_channel will be implemented in the experimental design as the hypothesized primary source of variation.

Minimal transformation is required for the target metric, `is_canceled`, as it is already a binary variable.

Notable covariates for potential confounding are lead time, deposit type, hotel, customer type, total of special requests. These were chosen due to domain knowledge and exploratory inspection. Several graphs were generated to examine the relevant distributions.

There is sufficiently large sample size for groups (in thousands) for Central Limit Theorem and logistic model convergence. It can also be safely assumed that bookings are independent across observations.

Experimental Design

The treatment is framed as the booking_channel while the response variable as is_canceled.

The target metric is the cancellation status, defined by the variable `is_canceled`, where a 1 = canceled booking and 0 = completed booking. This variable is binary and is therefore assumed to follow a Bernoulli distribution at the individual level. When aggregated across a group of observations, the number of cancellations follows a Binomial distribution.

To validate the use of normal approximation methods (e.g., the two-proportion z-test), we verify that the sample sizes are large and that both $n \times p$ and $n \times (1 - p)$ exceed 10. In our case, both groups are large (in thousands). As a result, the Central Limit Theorem justifies using normal approximations for inference on group proportions.

We use the two-proportion z-test to determine whether there is a statistically significant difference in cancellation rates between online and offline bookings. Null hypothesis (H_0)= the cancellation proportions are equal. The alternative hypothesis (H_1) =the proportions are not equal.

Later, after doing power analysis on the test, we elect to use Binary logistic regression to isolate adjusted treatment effect given confounding present. Power analysis and minimum detectable difference (MDD) calculations are made to understand the validity of our methods.

Experiment Results

Two-Proportion Z-Test
- *Proportion (online): 0.3672*
- *Proportion (offline): 0.3733*
- *z = -2.1659, p = 0.0303*
- *Small effect size (0.0061) with weak power*

Post Hoc Power
- *Calculated effect size: 0.0105*
- *Harmonic mean sample size: 59,623*
- *Power: 33.8% (low)*
- *MDD at 80% power: 0.0157*

Logistic Regression Results (Final Model) *[odds ratio=OR]*
- *Booking channel (Online): OR = 3.55, 95% CI: [3.40, 3.71]*
- *Deposit type (Non Refund): OR ≈ 0.003*
- *Total special requests: OR = 0.48*
- *Lead time: OR = 1.005 per day*

Nested Model Effects
- *Model 1 (Channel only): OR = 0.97*
- *Model 2 (+ lead time): OR = 1.26*

- *Model 3 (+ deposit type): OR = 2.78*
- *Final model (all covariates): OR = 3.55*

Discussion & Summary

The two-proportion z-test revealed a small but statistically significant difference in cancellation rates: 36.7% for online and 37.3% for offline bookings ($z = -2.1659$, $p = 0.0303$). However, the effect size (0.0061) was minimal, and post hoc power analysis showed only 33.8% power, with a minimum detectable difference of 0.0157 at 80% power. This suggests the raw difference lacks practical significance.

To better estimate the treatment effect, logistic regression was employed. After adjusting for confounders, online bookings were associated with over three times the odds of cancellation (OR = 3.55, 95% CI: [3.40, 3.71]). Bookings with non-refundable deposits had significantly lower cancellation odds (OR ≈ 0.003), and higher special request counts were also linked to reduced cancellations (OR = 0.48). Lead time showed a modest positive association (OR = 1.005/day).

Nested models showed how the apparent null effect (OR = 0.97) became substantial as covariates were added, reinforcing the impact of confounding. The final model demonstrated that much of the treatment effect was obscured in unadjusted comparisons, especially by deposit type and customer intent.

These findings underscore the operational importance of booking policies. Deposit requirements and engagement indicators like special requests may serve as effective tools to reduce cancellations, especially for online bookings.

Limitations

This study relies on observational data, which limits causal inference. Key logistic regression assumptions, such as logit linearity and absence of multicollinearity, were not formally tested. The sample is also geographically limited to two Portuguese hotels, which may affect generalizability. Unobserved variables like pricing or loyalty status were not included and could influence results.

Conclusion

While unadjusted differences in cancellation rates appear minor, adjusted analysis reveals that online bookings are significantly more prone to cancellation. This suggests a need for stricter cancellation policies or customer engagement strategies in online channels. The study also highlights the importance of proper covariate control and power analysis in observational designs, especially when experimental data are unavailable. Online bookings, while convenient and widespread, may require more robust cancellation policies or incentives to reduce no-shows.

From a methodological standpoint, the study also underscores the importance of simulating experimental design principles—such as covariate adjustment and power analysis—when analyzing observational data. Though randomized experiments remain the gold standard, carefully structured statistical analyses can yield valuable insights when experiments are infeasible.